FOR A UNIVERSALLY CLOSED SOURCE MAP THE LOCUS WHERE A MORPHISM OF SPACES IS A CLOSED IMMERSION IS OPEN

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 11641-11659 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 11661-11684, the complete author proof (22 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The representability-by-open-subspace phrasing is the functor-of-points formulation of openness of the closed-immersion locus.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The closed-immersion characterization lemma of the same chapter (collected as L16040), the open-locus result for unramified universally injective maps (lemma-where-unramified-universally-injective), and permanence of universal closedness (Morphisms of Spaces).

Source reference. Original lemma 05W8: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2906–2917. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05X9: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11566–11579. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03IS: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11684–11684. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03IU: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11684–11684. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04NX: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11684–11684. Cited by the retained statement or

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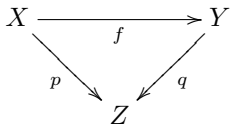
proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Consider a commutative diagram*



of algebraic spaces. Assume that

(1) *p is locally of finite type,*
(2) *p is universally closed, and*
(3) *$q : Y \to Z$ is separated.*

Then there exists an open subspace $W \subset Z$ such that a morphism $Z' \to Z$ factors through W if and only if the base change $f_{Z'} : X_{Z'} \to Y_{Z'}$ is a closed immersion.

Proof. We will use the characterization of closed immersions as universally closed, unramified, and universally injective morphisms, see Lemma 05W8. First, note that since p is universally closed and q is separated, we see that f is universally closed, see Morphisms of Spaces, Lemma 04NX. It follows that any base change of f is universally closed, see Morphisms of Spaces, Lemma 03IS. Thus to finish the proof of the lemma it suffices to prove that the assumptions of Lemma 05X9 are satisfied. The projection $\mathrm{pr}_0 : X \times_Y X \to X$ is universally closed as a base change of f, see Morphisms of Spaces, Lemma 03IS. Hence $X \times_Y X \to Z$ is universally closed as a composition of universally closed morphisms (see Morphisms of Spaces, Lemma 03IU). This finishes the proof of the lemma. □